Why bigger motors lift all three, and why the ladder is brass

The working behind the lift revision approved on 2026-11-10: lift bought with disc as much as motor, what a 250 mAh pack can feed and why the throttle is capped, the brass ladder and its stiffness, a 6 mm lever that nods the camera 30°, rangers that sit low and look between the discs, five sensors with one address behind one switch, and where the weight went. Read after XS Drone Custom Build and XS Drone Voice Physics. Nothing measured yet: every figure is derived from the listings, the datasheets and the build page's numbers; the motor thrust is a range from physics, since no vendor sheet could be read. Source of truth: `xs_drone/docs/lift-physics.md`.

The whole thing in one breath

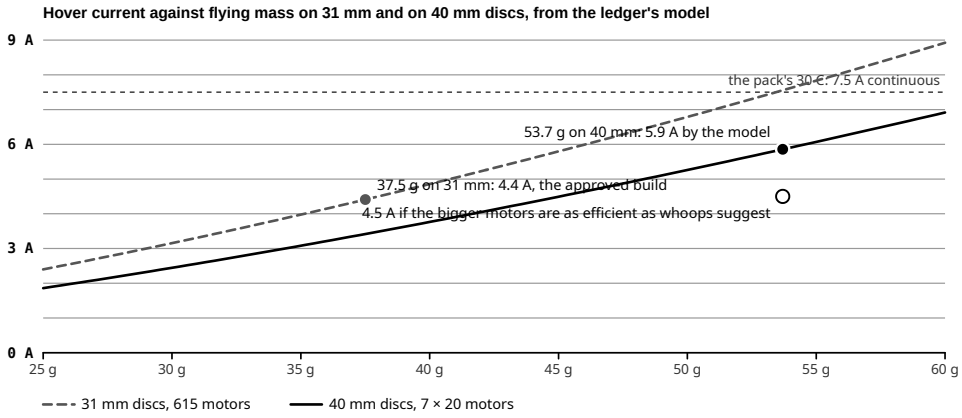
Three things were asked of the drone at once: a voice, a camera that nods, and eyes all round. Together they weigh 16.2 g more than the 37.5 g the approved build lifted at 1.81 to one, and 53.7 g on 68 gf would hover at 80 % throttle with nothing in reserve. The cheapest lift in the world is the whoop world's own next size: a motor a size larger on a propeller a size larger, nothing else electrical. Four 7 × 20 mm motors on 40 mm discs give, by the physics below, 121 to 148 gf, 2.2 to 2.8 to one, for about 7 g of motor, prop and longer axle.

The rest of this page is why each piece of the revision is shaped the way it is: why the disc matters as much as the motor, what a 250 mAh pack can and cannot feed, why the ladder went to brass and what that costs in stiffness, how a 6 mm lever turns 6 mm of travel into 30° of nod, why the rangers sit low and look sideways between the discs, and why five identical sensors need one small switch and no pins.

Lift is bought with disc, not just with motor

A hovering rotor throws air down, and the power it needs to hold a thrust T is T to the power 1.5 over the square root of twice the air density times the disc area. Two things follow. Doubling the mass costs 2.8 times the power, which is why every gram shows in the flight time. And at a fixed weight, a bigger disc lifts it more cheaply: the 40 mm disc has 1.66 times the area of the 31 mm one, so the ideal power falls by the square root of that, 78 %. Momentum theory gives 3.4 W of ideal power for 53.7 g on four 40 mm discs; the approved build's 37.5 g on 31 mm discs needs 2.6 W ideal and about 16 W electrical, so the little brushed motors and their propellers turn about 16 % of the pack's power into lift. That is normal for this size.

The thrust a motor can make follows from the power it can put in: at a fixed efficiency, thrust goes as the two-thirds power of power times the square root of disc area. A 7 × 20 motor takes about 3 A at full throttle against the 615's 1.2, which with the bigger disc multiplies the thrust by 2.2: 148 gf from the 68. Taking the motor at 2.2 A instead gives 121. No vendor publishes a thrust figure for these motors, so the build document carries 120 to 140 gf as the working range, this page 121 to 148 from the physics, and the first hover reads the truth, as it always did; the gate stays 2.0 and the floor 1.5.



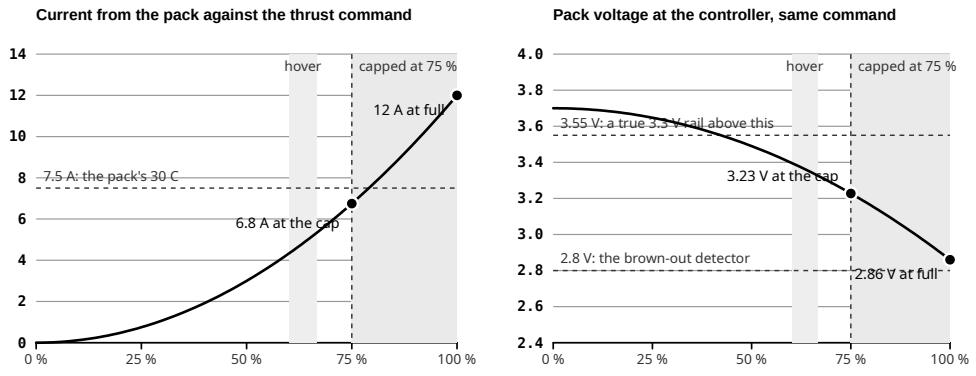
Hover current against flying mass on the two disc sizes, from the ledger's model: 2.4 A at 25 g on 31 mm discs, scaled as mass to the 1.5 and as the square root of disc area. The dashed line is the pack's 30 C. The 53.7 g lift revision sits at 5.9 A by the model and nearer 4.5 if the bigger motors are as efficient as whoop pilots report; the page uses 5 A.

The model is honest about what it knows: it is the ledger's 2.4 A at 25 g carried to the new mass and the new disc. It does not know that a 7 × 20 motor wastes less of its current in brush and winding losses than a 615, which the whoop world's flight times say it does. So the hover current for 53.7 g is 4.5 to 5.9 A, about 5; the build document says the same. At 5 A the pack is at 21 C, and 80 % of 250 mAh less 5 % for the controller and camera is 2.2 minutes in the air, 1.9 to 2.5 across the range.

What a 250 mAh pack can feed, and why the firmware caps the throttle

A brushed motor on a propeller draws current about as the square of its throttle command: the speed follows the voltage, the load follows the speed squared. Four 7 × 20 motors at full throttle take about 12 A. The pack is rated 30 C, 7.5 A continuous, and a small cell behind a PH 2.0 pigtail has about 70 mΩ in its way, so 12 A would pull its terminals down to 2.86 V. The controller's 3.3 V regulator wants about 3.55 V at its input for a true 3.3 V, and the chip's brown-out detector trips near 2.8 V: a full-throttle punch on this pack would reset the flight controller in the air.

So the firmware caps the thrust command at 75 %. At the cap the motors draw 6.8 A, inside the pack's 7.5, and the terminals sit at 3.23 V; the hover command for 2.2 to 2.8 to one is 60 to 67 %, well under the cap, with the full margin between still there for a gust. The thrust gate is read by the hover command, not by a punch, which is how D02 was written in the first place.



Left: current from the pack against the thrust command, rising as the command squared, past the pack's 7.5 A at about 79 %; the hover band and the 75 % cap are shaded. Right: the pack voltage the controller sees, falling with that current through 70 mΩ, against the regulator's need and the brown-out line. Resistances are assumptions, named at the top of the script.

The way to more margin is a bigger pack, not a bigger C rating: a 450 mAh pack has twice the plates, half the resistance and twice the flight time for about 5 g, but it is 17 mm wide and the pan between the rails is 14. That is a later revision, not this one. For now the 220 µF across the switch cells and the 47 to 100 µF at the amplifier hold the rails through each pulse, and the cap keeps the controller awake.

A brass ladder, and what it costs in stiffness

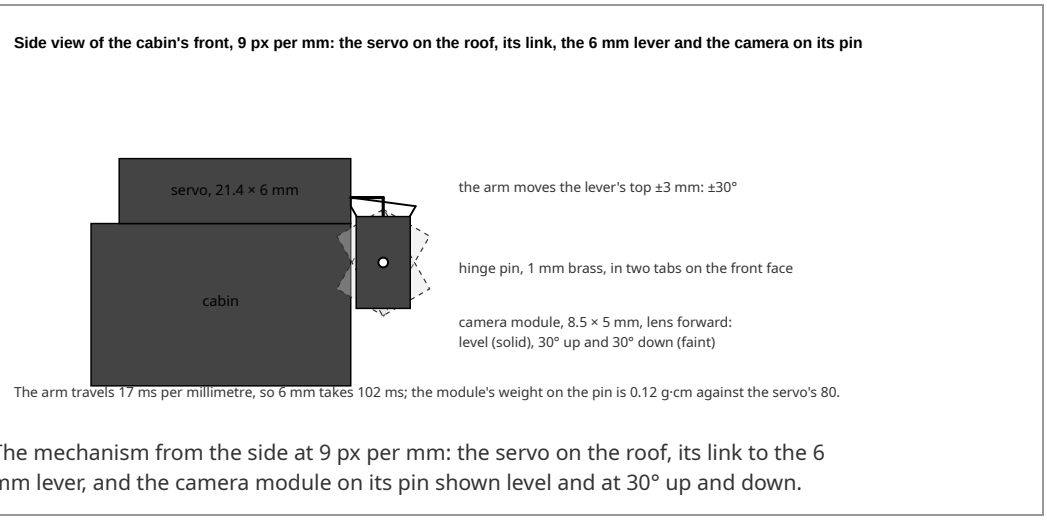
The builder asked that the drone match the desk family's brass, and nothing is painted, so the only brass a whoop can wear is its frame. The ladder is 305 mm of 1.0 mm brass rod and 212 mm of 1.5 mm for the four collars: 5.2 g against 4.9 in stainless, 7 % more, because brass is 8.5 g/cm³ to stainless steel's 7.93. The rod is already in stock, so the frame costs nothing to buy.

Brass is about half as stiff: Young's modulus 100 GPa against 193. A rail is a 1 mm beam spanning 60 mm between the axles and carrying half of the 15 g of pan, cabin and roof; with the cubic law of beam bending it sags 67 µm in brass and 35 in stainless, and 169 µm in brass at the 2.5 g of a hard punch. A quarter of a millimetre on a frame that flexes its rubber props more than that is nothing, and in any case the strapped pack turns the two rails into the flanges of a 6 mm-deep beam, many times stiffer than the rods alone. Stiffness was never the reason for stainless; its hardness and its indifference to tarnish were.

Soldering is where brass wins. Copper alloys carry a thin oxide that rosin flux reduces, so a brass joint wets with ordinary tin and leaves a residue that needs only a wipe of alcohol. Stainless hides under a chromium oxide skin that only an acid flux breaks, and that flux must be washed off or it keeps eating. Brass also conducts heat 7 times better than stainless, which means a joint drains the iron faster: a larger tip and a moment longer on the rod, and the collar former left in while the collar is soldered so it does not close. Brass tarnishes toward brown over months, which is in the family's palette and is left alone.

A 6 mm lever: how 6 mm of travel becomes 30° of nod

A quadcopter cannot hold a tilted hover, so the camera is hinged and the drone's body stays level. The module hangs on a 1 mm brass pin through two tabs on the cabin's front face, with a 6 mm lever standing up from the pin. The servo lies on the roof with its arm forward and a 1 mm brass link to the lever's top. A linear servo moves its arm in a straight line, and a lever's top moved sideways by s swings the lever through the angle whose sine is s over the lever's length: 3 mm on 6 gives 30° each way, using 6 of the arm's 9 mm.

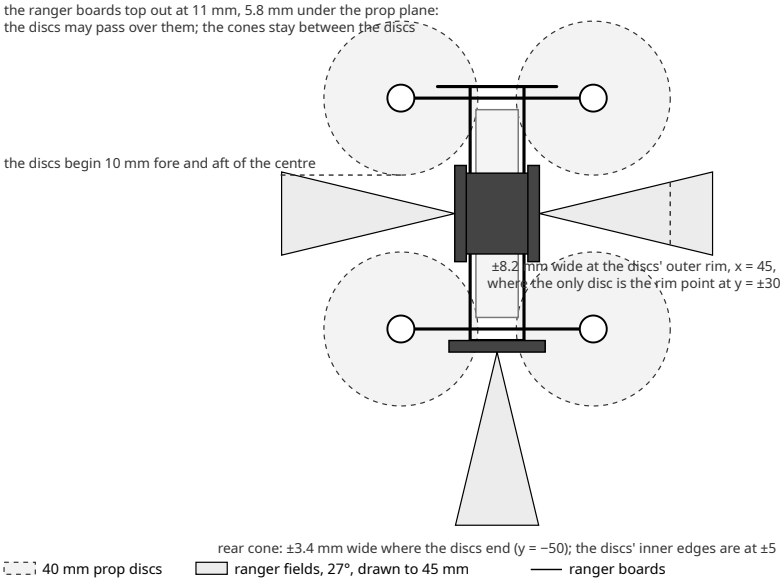


The mechanism from the side at 9 px per mm: the servo on the roof, its link to the 6 mm lever, and the camera module on its pin shown level and at 30° up and down.

The load is almost nothing: the module weighs 0.5 g with its centre about 2.5 mm ahead of the pin, 0.12 g·cm, against the servo's rated 80; the flex cable's stiffness is the real resistance, and a servo made to move a model aeroplane's elevator in flight does not notice it. The arm moves 17 ms per millimetre at 3.7 V, so a nod from one stop to the other takes about 102 ms. The signal is the ordinary 50 Hz pulse of 1 to 2 ms, which the controller makes with its LED timer on the pin the multiplexer freed; the servo's electronics run from the pack and read 3.3 V as high.

Eyes all round, low: why the rangers sit where they do

A single-zone ranger sees a cone 27° wide. Anything inside that cone that is nearer than the wall becomes the reading, and a propeller blade passing through it would be read as a wall 4 cm away. So the side rangers sit low, outside the rails at the cabin's foot, where their boards top out 11 mm above the rails, 5.8 mm under the prop plane: the discs may pass over them. Their cones look straight out to the sides from the middle of the body, and the discs begin 10 mm fore and aft of that line. At the discs' outer rim, 45 mm out, a cone is only ±8.2 mm wide, and the only part of a disc that far out is its rim at y = ±30. The cones and the discs never share a point.



The four discs, the three low ranger boards and their cones drawn to 45 mm, from above. The side cones pass between the front and rear discs; the rear cone leaves from behind the tail bar with nothing ahead of it.

The rear ranger moved from the cabin's back face to behind the tail bar for the same reason: on the back face its 25 mm board would have reached under the rear discs with its top above the prop plane, and it would have covered the controller's USB port. Behind the tail bar, at 36 mm, it looks back with nothing in its field at all, 5.8 mm under the discs, and the USB port stays free. The boards hang on two 1 mm brass posts or a brass tab and B-7000, and each board's shutdown pin is tied to its own supply so that none depends on a pull-up that may or may not be fitted.

Five sensors with one address, and one small switch

Every VL53 ranger wakes at the same I²C address. The usual way to put several on one bus is to hold all but one in reset at boot and give each a new address in turn, which costs a wire per sensor, and the controller has none left. The multiplexer is the other way: a chip with eight pairs of analog switches, one per channel, that connects exactly one pair to the bus when told to by a single byte written to its own address. Five rangers on five channels never see each other, so all five keep their default address and nothing is re-addressed. The motion sensor stays on the bus itself, at its own address, outside the switch.

The bus has room. Five rangers read 15 times a second at about 20 bytes a read are 12 kbit/s, the motion sensor at 500 Hz another 56, together about 21 % of a 400 kHz bus with its overheads. The switch adds a few nanoseconds and a few picofarads to each channel, and because only one channel is on at a time the bus never carries more than one ranger's capacitance. The board lies across the hood, chip down, 8 mm under the prop plane; it is the one new part that shows, and it is black.

Where the weight went, and the balance

The revision adds 16.2 g to the voice build: motors 6.0, the brass ladder 0.8 over stainless, props 0.8 (the grey polycarbonate HQ props weigh 0.5 g each, more than the 1636s first counted), multiplexer 1.8, three rangers and their posts 3.3, servo and hinge 2.5, wiring 1.0. The motors and props sit at the four corners and cancel; the multiplexer rides ahead of the centre and the speaker and rear ranger behind it, so the centre of gravity moves 0.24

mm toward the tail and the rear motors carry 50.4 % of the hover instead of 50. The controller trims that without being told.

At 53.7 g the margins are these: the 2.0 gate is met up to 60 to 74 g depending on where the thrust really lands, and the 1.5 floor up to 80 g at the low end. If the first hover reads under 2.0, the build document names the lighter paths; if it reads over 2.4, the margin is a 450 mAh pack and twice the time in the air.

The numbers, and where they come from

NUMBER	WHAT IT IS
1.66	the disc area ratio: $(40/31)^2$; ideal hover power falls by its square root, 78 %
3.4 W	ideal hover power for 53.7 g on four 40 mm discs: 4 × T^1.5 / √(2ρA), T = weight/4
16 %	the approved build's ideal power over its electrical power: 2.6 W over 16
121 to 148 gf	thrust from power scaling: 68 gf × (I_{720}/I_{615} × √area ratio)^(2/3), with the 7 × 20 at 2.2 to 3 A against the 615's 1.2
2.2 to 2.8	thrust to weight at 53.7 g; hover command √(1/TW) = 60 to 67 %
4.5 to 5.9 A	hover current: the ledger's model gives 5.9, whoop experience nearer 4.5; 21 C at 5 A
2.2 min	in the air: 80 % of 250 mAh over 5 A, less 5 %; 1.9 to 2.5 across the range
12 A, 6.8 A	four motors at full throttle, and at the 75 % cap (current as command squared); the pack gives 7.5
2.86 V, 3.23 V	pack terminals at full and at the cap: 3.7 V less the current through 70 mΩ; the regulator wants 3.55, brown-out near 2.8
5.2 g	the brass ladder: 305 mm of 1.0 mm and 212 mm of 1.5 mm rod at 8.5 g/cm³ (4.9 g in stainless)
67 µm	a rail's sag under half of 15 g over 60 mm: F L^3 / 48 E I with E 100 GPa (35 µm in stainless; 169 at 2.5 g)
30°	the camera's swing each way: arcsin(3 mm / 6 mm lever)
102 ms	stop to stop: 6 mm at 17 ms/mm
0.12 g·cm	the module's moment on the pin: 0.5 g × 2.5 mm; the servo is rated 80
±8.2 mm	a side cone's half-width at the discs' outer rim: (45 − 11) mm × tan 13.5°; the discs begin 10 mm fore and aft
5.8 mm	the ranger boards' tops under the prop plane: 16.8 − 11
21 %	of the I^2C bus: five rangers at 15 Hz × 20 B plus the motion sensor at 500 Hz × 14 B, on 400 kHz at 80 %
0.24 mm	the centre of gravity's shift to the tail from the parts off the centre line; the rear motors carry 50.4 %

Which part each idea lives in

- **lift from disc as much as motor:** 7 × 20 mm coreless motors, 4-pack with JST 1.25 leads, ordered 2026-11-10
- **the 40 mm disc, grey polycarbonate:** HQ Prop 40 mm four-blade, 1 mm shaft, two sets in the tinywhoop.com cart
- **7.5 A and 70 mΩ: the cap at 75 %:** the 250 mAh 30C flight packs on hand

- **half the stiffness, rosin flux, brown with time:** K&S 1.0 mm brass rod on hand
- **the four collars:** K&S 1.5 mm brass rod on hand
- **9 mm of stroke, 6 used, 30° each way:** AGFRC 1.5 g linear servo pair, ordered 2026-11-10
- **eight switches, one byte, no pins:** Adafruit TCA9548A multiplexer, ordered 2026-11-10
- **27° cones that never meet a disc:** TOF400C VL53L1X boards, AITRIP 2-pack ×2, ordered 2026-11-10
- **the cutting sheet for the brass:** the frame jig, regenerated for 50 mm axles and 7 mm collars

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What comes next

When the parts come: weigh the motors, props, multiplexer, boards and servo on the coin balance and correct the ledger; measure one 7 × 20 motor's current at a hover duty and at full on the bench supply (test D01 and the switch cell's diode warmth) and set the cap before the first hover. Read the thrust gate from the hover command: under 67 % means at least 2.2 to one. Tests C13 (every ranger answers on its channel and none reads a propeller with the props spinning on the bench) and C14 (±30° on a protractor) close this revision. The following page still draws the 37.5 g build; it gets the new geometry when the brass frame is soldered and weighed.

Sources

- Momentum theory of a propeller, NASA Glenn
- TCA9548A datasheet, Texas Instruments
- VL53L1X datasheet, STMicroelectronics
- Young's modulus of common metals, Engineering ToolBox
- Thermal conductivity of metals, Engineering ToolBox
- The motor, prop, servo, multiplexer and ranger listings read at the cart on 2026-11-10 (links above)
- XS Drone Custom Build, the sheet this page explains
- XS Drone Voice Physics, the previous step-two page

Built by `docs/pages/gen/lift_page.py`, which also writes `docs/lift-physics.md`. Assumptions, each a number at the top of the script: the 615 at 1.2 A and the 7 × 20 at 2.2 to 3 A at full throttle; the ledger's hover model of 2.4 A at 25 g; 40 mΩ in the cell and 30 in the pigtail; a regulator that wants 3.55 V and a brown-out near 2.8; 15 g on the rails; a 0.5 g module 2.5 mm ahead of its pin; 20 bytes a ranger read. Brass and stainless figures are handbook values; the servo's are its listing's. Nothing here has been measured on the drone.